

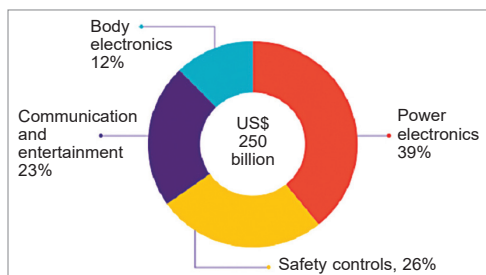
The automotive electronics iceberg

The Automotive Component Manufacturers Association (ACMA) of India recently released a report titled 'Auto-Electronics Manufacturing: Conquering the Next Frontier.' Conducted with Grant Thornton, the report identifies electronic automotive components as a market opportunity worth over \$540-650 billion by 2032.

Products like reverse parking guide, DC-DC converter, charger system, tire pressure monitoring system, controller, telematics unit, navigation system, electronic power steering, power distribution module, thermal management system, and engine control system are highlighted for their high potential and ease of manufacturing. Coupled with nine more products, these are projected to grow from a current \$9.3 billion opportunity to \$56.4 billion in 2032.

Vineet Sahni, CEO and Director of FIEM Industries, informs EFY, "The electronic content in cars has grown 20 times. The rate of change has been exponential and is still increasing."

On a global scale, the automotive electronics market is expected to grow at a CAGR of 8–10%, reaching \$540-650 billion in 2032 from \$250 billion in 2022. According to Statista, India's automotive parts exports were valued at around \$19 billion in the financial year 2022, marking a significant increase from the previous year. Europe and North America were the primary export markets for India's auto components industry.



Breakdown by global auto-electronics demand by product categories, 2022 (Source: ACMA)



Infotainment screen of Yamaha R15: new (left), old (right)

Additionally, the increasing number of electric vehicles (EVs) presents an opportunity. EVs share many components with their internal combustion engine (ICE) counterparts, including infotainment systems, LED lights, window glass motors, and more.

According to MG's Manek, "These are excellent opportunities for electronics component makers based in India. We have witnessed traditional component makers, previously strong in the mechanical components vertical, diversifying into electronic components."

He adds, "In ICE cars and EVs, several components come from the same family. This represents a natural extension for an existing component maker."

Even in the latest electric scooter launch from TVS, the focal point was not its powertrain, battery size, or futuristic looks, but the 26cm tilt screen loaded with infotainment features. Similarly, Yamaha, Honda, Suzuki, Bajaj, and Hero have begun offering LED displays as infotainment screens. Previously, these were only available in high-end motorcycles in developed markets like Europe and the United States.

The lowest hanging fruits

India, as confirmed by ACMA, heavily relies on imports to meet the demand for auto-electronic products, with around 64% of the total demand being im-

ported. Furthermore, the domestic market for automotive electronics (OEM + Aftermarket) is expected to reach \$63.7 billion by 2032.

While electronics used in exhaust gas circulation, electronic power steering, airbag electronics, anti-lock braking systems, and fully automatic temperature control are in high demand, the electronic content used in safety controls currently has the highest demand in the country.

This is followed by power electronics, communication electronics in entertainment consoles, and body electronics. Within power electronics, power electronics of internal combustion engine cars currently dominate the market.

Bohra tells EFY, "Almost every component in a vehicle now has a sensor. The centre console is completely electronic, and the instrument cluster is electronic; many cars now display driver information electronically. Even smaller components such as window switches, gloveboxes, switches, mirror door switches, and more are becoming electronic. ADAS and telematics are becoming significant in cars, and car brands increasingly want these features in their vehicles."

He adds, "A plethora of electronics is entering the ICE vehicle industry. In fact, we often jest cars, once primarily regarded as mechanical components, are now electro-mechanical equipment."

He also notes that depending on the segment, there may be additional